

DDR5 Temperature Sensor (TS5)



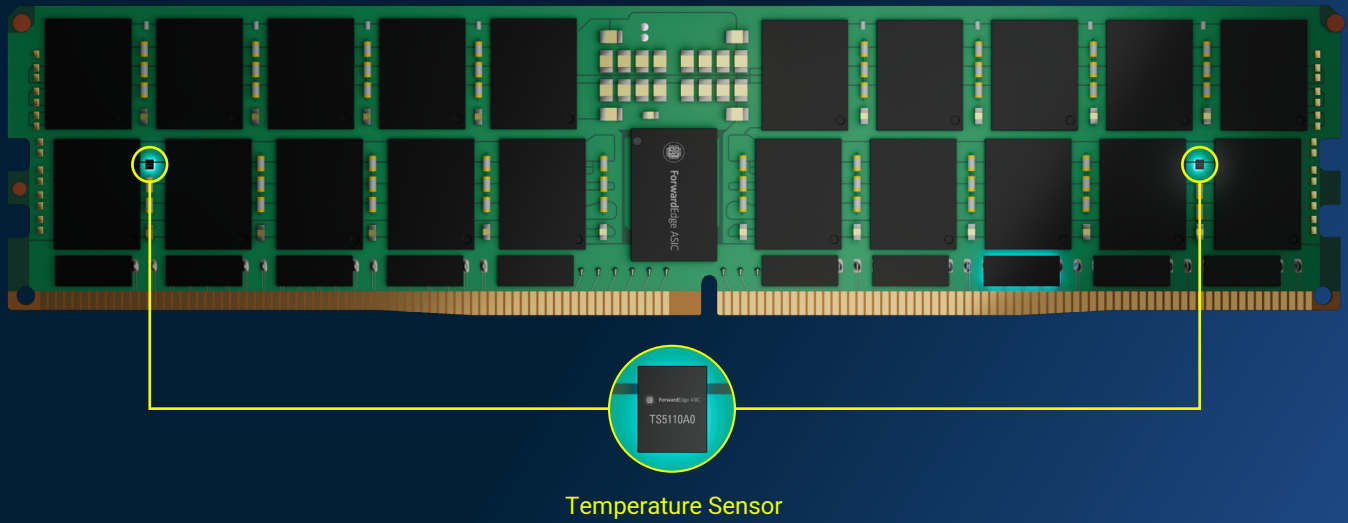
Product Description

TS5110A0 / TS5210A0 device incorporates thermal sensing capability which is controlled and read over two wire bus. TS5 device operate from a nominal 1.8 V nominal power supply (VDDSPD) and a 1.0 V nominal power supply (VDDIO).

The TS5 device is intended to operate up to 12.5 MHz on an I3C Basic bus or up to 1 MHz on an I2C bus. The TS5 devices are intended to interface to I2C or I3C Basic buses which have multiple devices on a shared bus and must be uniquely addressed with fixed addressing on the same bus. All TS5 devices respond to specific pre-defined device select codes on the two-wire bus.

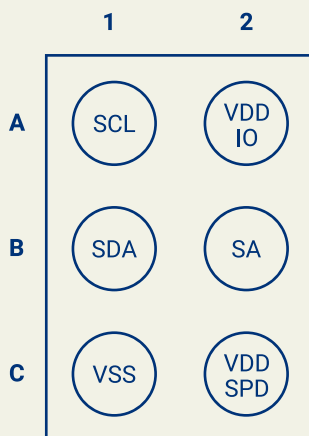
Feature List

- Two wire bus serial interfaces (I2C and I3C Basic operation modes)
- Up to 12.5 MHz transfer rate
- 6 ball WLCSP Package device Functional Diagram
- 1.8 V & 1.0 V power supply input
- Temperature Sensor with high accuracy for JEDEC specification compliance
- Temperature Range from -40°C to +125°C
- Packet Error Check (PEC) Function
- Parity Error Check Function
- Bus Reset Function
- Two unique addresses selected by SA pin
- In Band Interrupt (IBI)
- 5-byte unique serial number to support authentication (For TS5210A0 only)



Package Pinout

The TS5 device is packaged in 6 ball WLCSP. Ball pitch is 0.5 mm. Package size is $0.8 + 0.05 \text{ mm} \times 1.3 + 0.05 \text{ mm}$. The ball diameter is 0.22 mm (after ball attach reflow) and maximum package thickness including the bump is 0.6 mm.



Applications

- DDR5 DIMMs for servers (RDIMM, LRDIMM and MRDIMM)

Contact Information

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